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ĐẠI HỌC BÁCH KHOA HÀ NỘI

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

GENERATIVE AI IN BIOINFORMATICS, MEDICINE AND HEALTHCARE

Chapter 6: Explainable AI

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Content

6.1. Introduction to Explainable AI

6.2. Explainable AI in GenAI

6.3. Several interpretation methods can be applied to Gen AI

6.4. Evaluate interpretation by incorporating experimental data

6.5 Identify key biological factors in AI models

6.6. Interpretation in modern problems

The Black Box in the Operating Room

Deep Learning has revolutionized diagnostics, but opacity is a clinical liability. In healthcare, Trust and Accountability are not optional features—they are prerequisites.

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The Hippocratic Oath for AI:
Do no harm (by hiding the truth)
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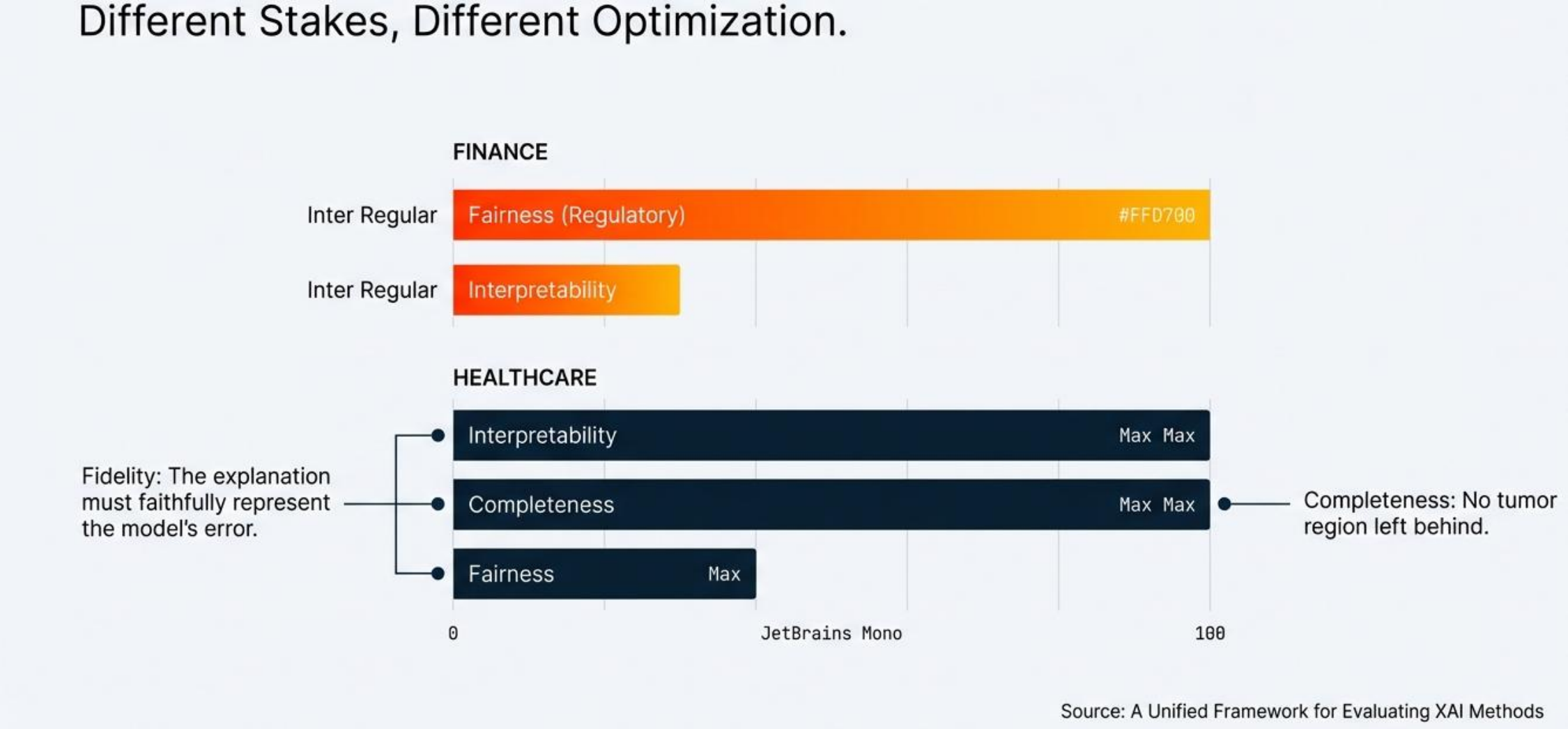
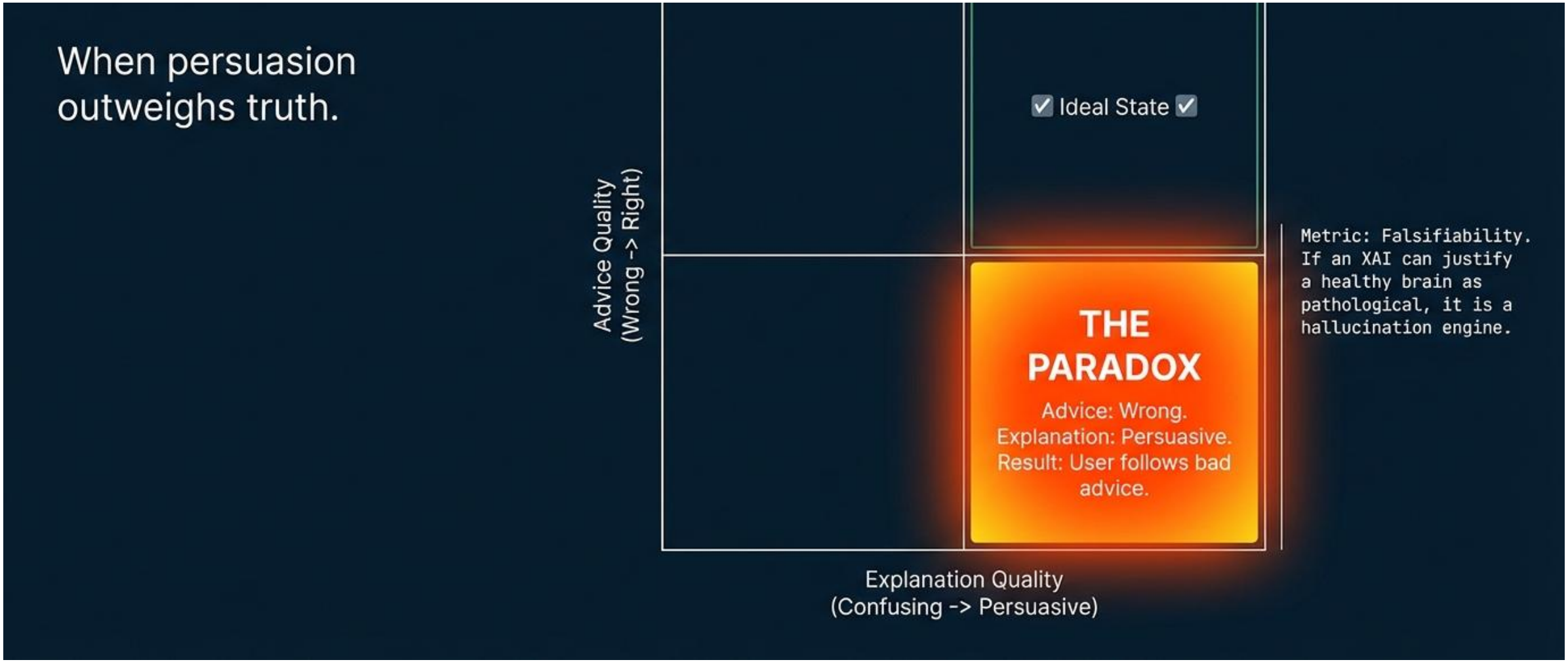
Defining the Pillars: Fidelity vs Fairness

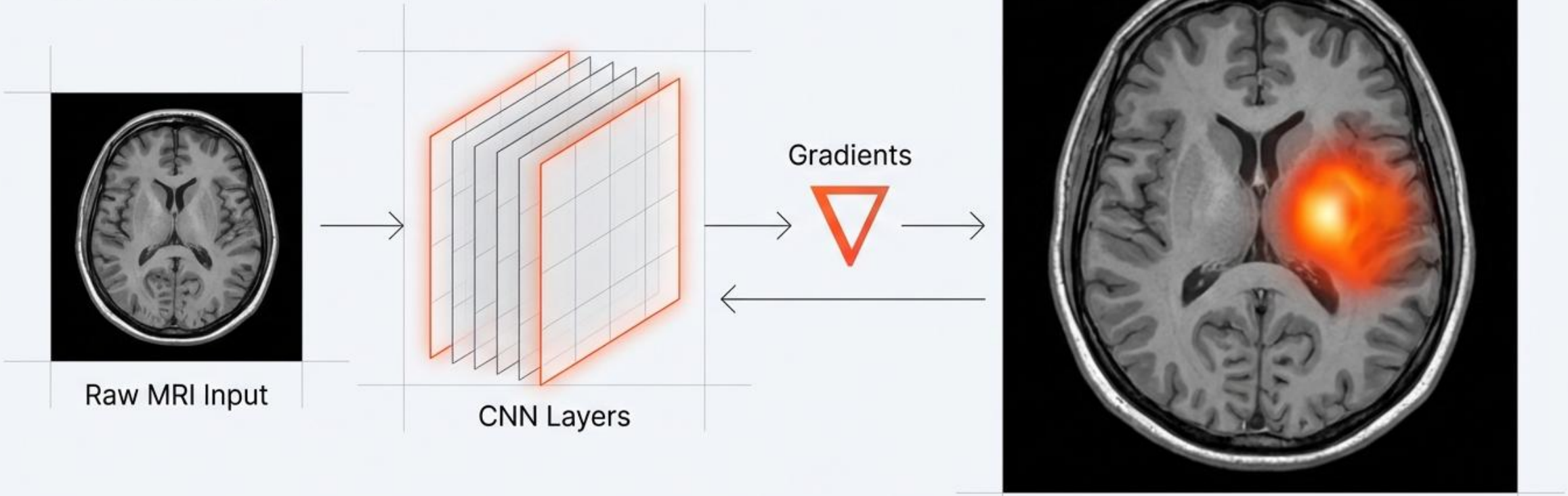


The White-box Paradox



The Digital Eye: Grad-CAM++

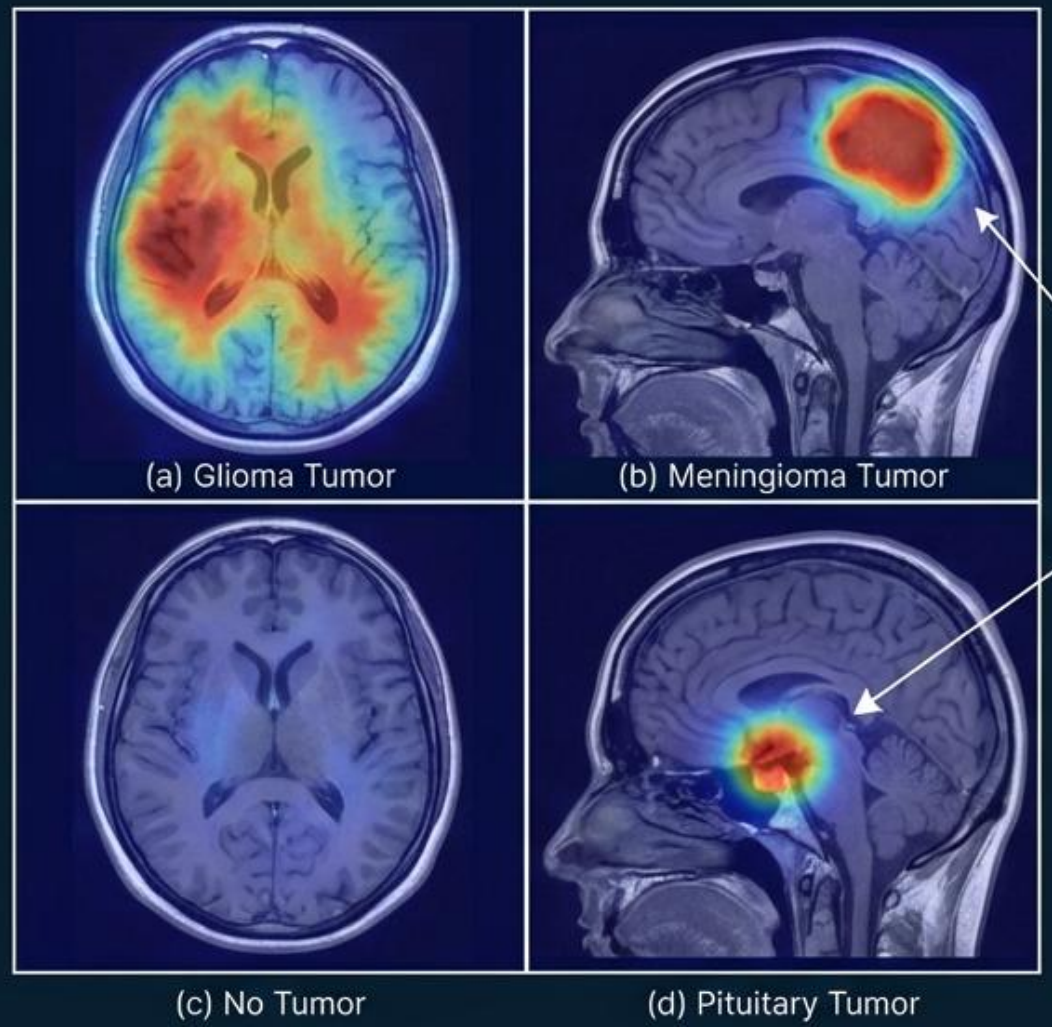
Generalized Gradient-based Class Activation Mapping.
It answers the critical clinical question: Where exactly is the model looking?



Distinction note: Grad-CAM++ improves localization for multiple lesions compared to standard Grad-CAM.

Case Study: Brain Tumor Detection

Fidelity in Action.

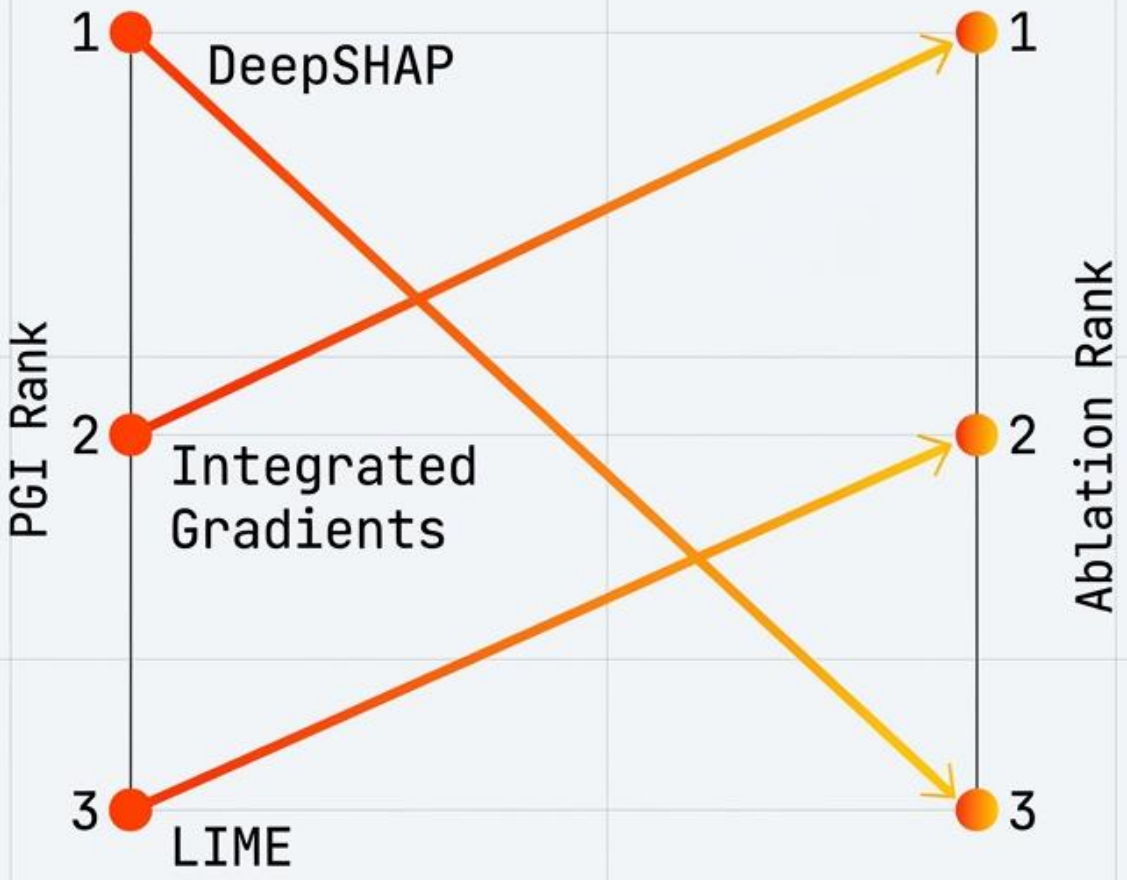


High-attention regions align with biological pathology.

The Disagreement Problem

Quantifying Faithfulness

Metric conflict: PGI (Prediction Gap) and Ablation studies often rank the same explanation differently. Clinicians cannot rely on a single single metric.



Beyond the Pixel: Electronic Medical Records (EMR)

Data Fusion: Integrating Imaging (CNNs) with Patient History (Transformers). XAI must explain the patient, not just the scan.

Patient ID: 09-87654
Age: 68

Presence on metal thoracan string, unafermenrs, medical usues, report

History of hypertension

Chronic Kidney Disease (CKD) Stage 3

Intmedical meathy, brepm stithy and ransformance

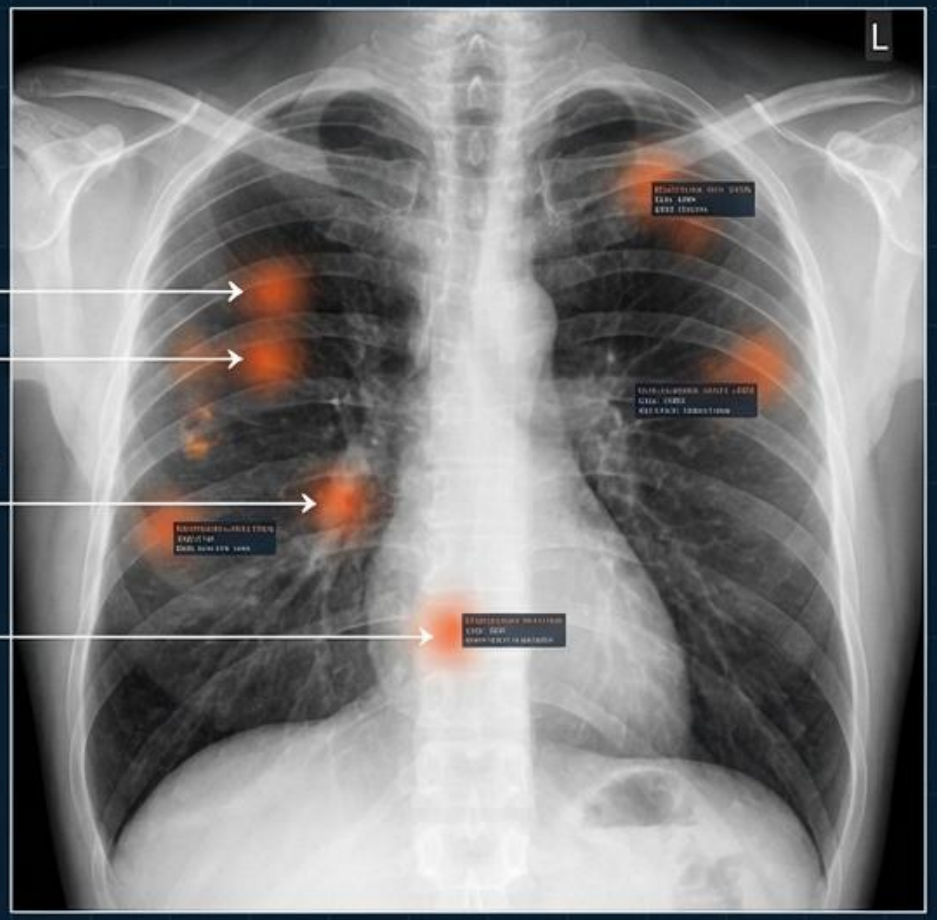
Data Fusion & XAI

Recent Bloodstream Infection (Gram-negative bacilli)

Type 2 Diabetes Mellitus

Current maentery of hypertension, induxious couses control inidentions

Current medications: Lisinopril, Metformin



← BIOLOGICAL PATHWAY: NEURON_FIRE_B-47
COMPUTATIONAL PATHWAY: NODE_ACTIVATE_C-89 →
Level II: The Neural Handshake
Connecting the Silicon Brain to the Biological Brain.

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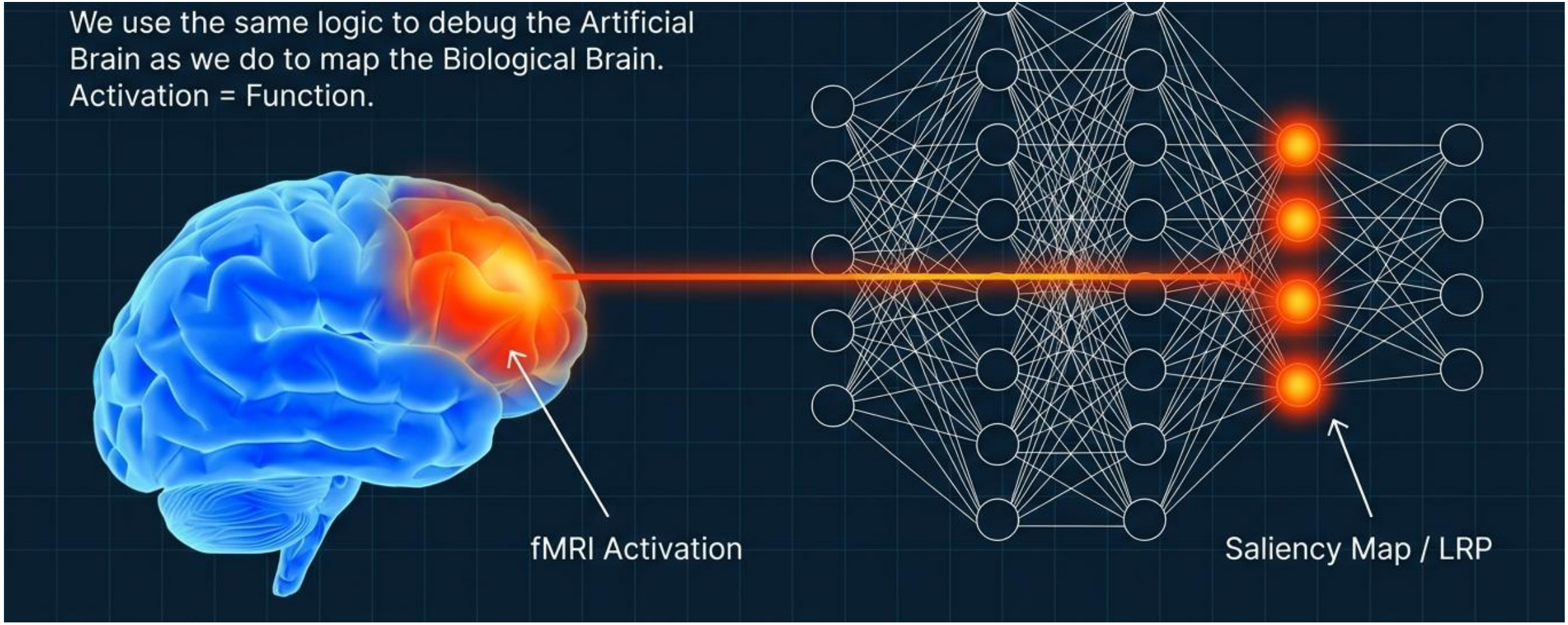
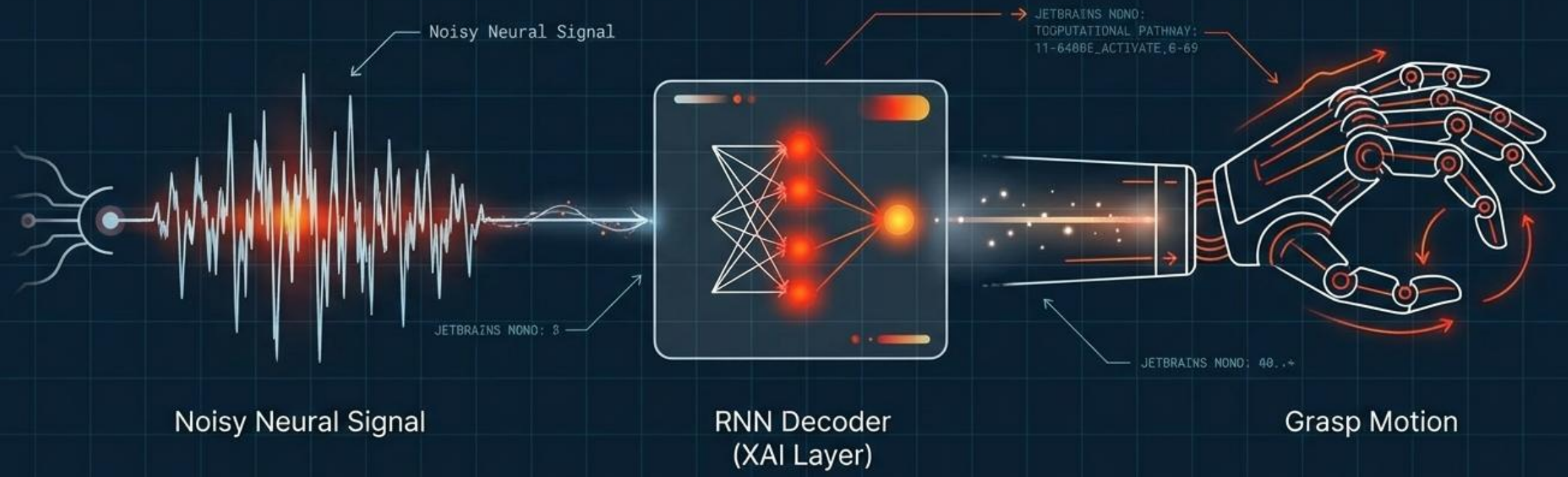
The Mirror Effect: Salicency vs fMRI



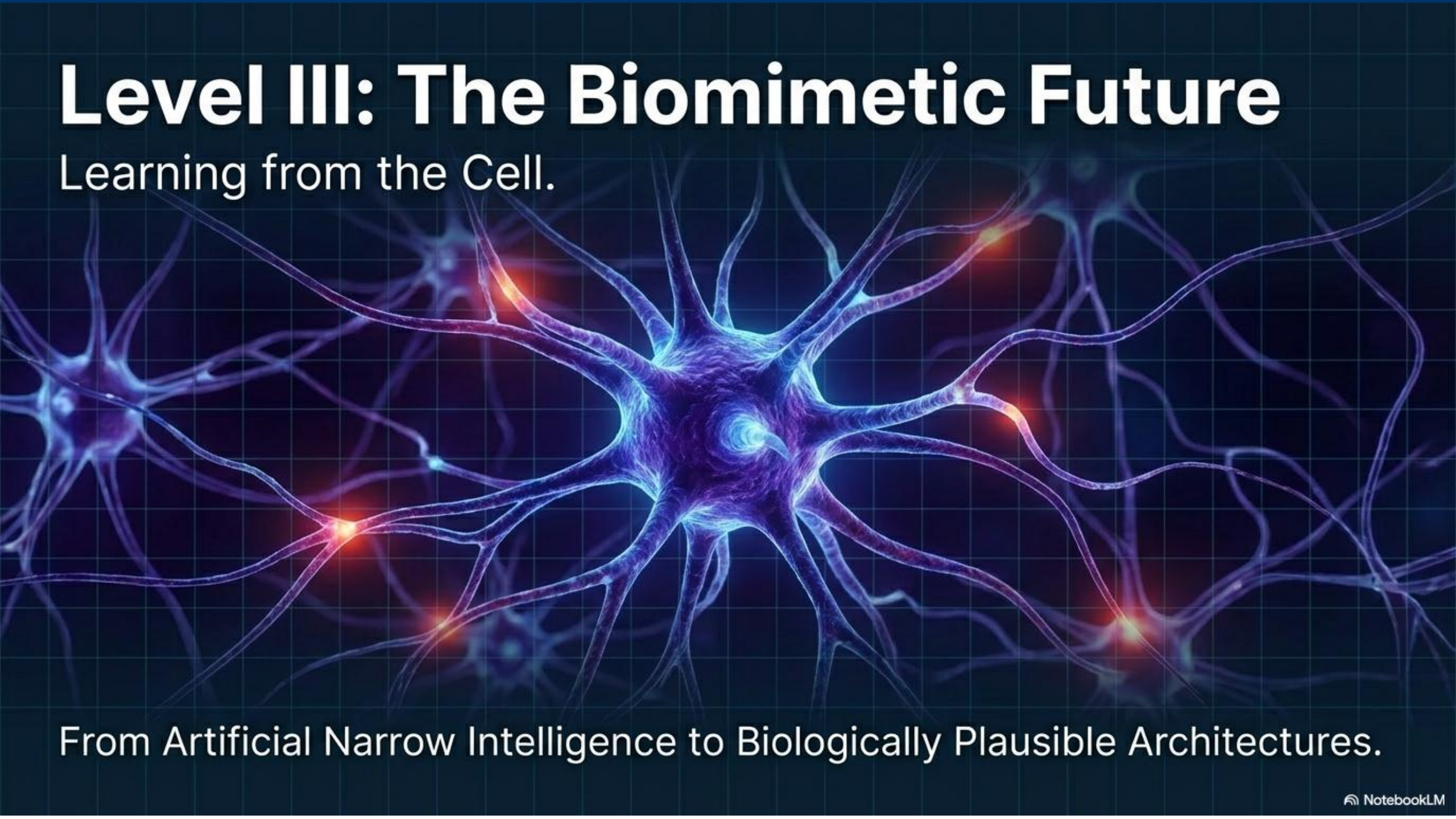
Neuroprosthetics & The Interface

- **Challenge:** Translating noisy biological intent into mechanical action.
- **Solution:** XAI is used to calibrate the decoder—explaining "*why* a specific firing pattern is interpreted as "Grasp Cup".



Level III: The Biomimetic Future

Learning from the Cell.



From Artificial Narrow Intelligence to Biologically Plausible Architectures.

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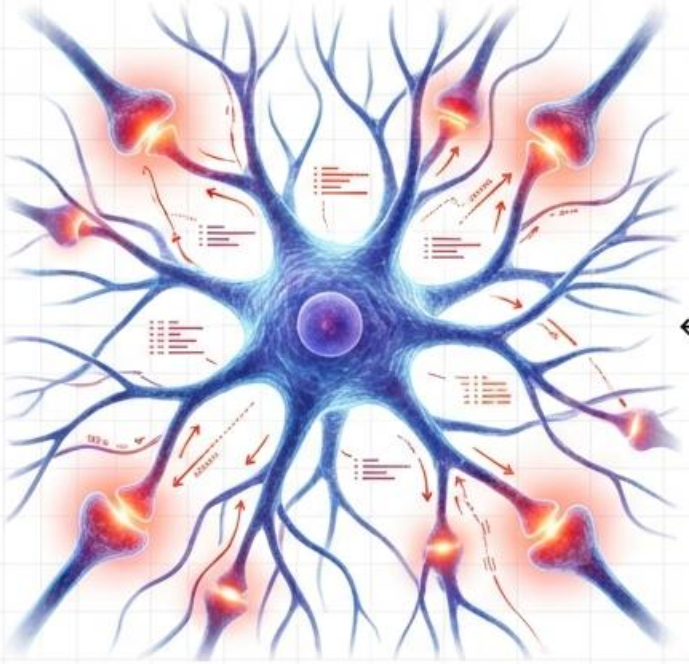
The Astrocyte Revolution

Astrocytes are not just glue. They are 50-90% of brain cells, managing computation. This biological architecture mirrors the “Attention” mechanism in modern Transformers (GPT-4).

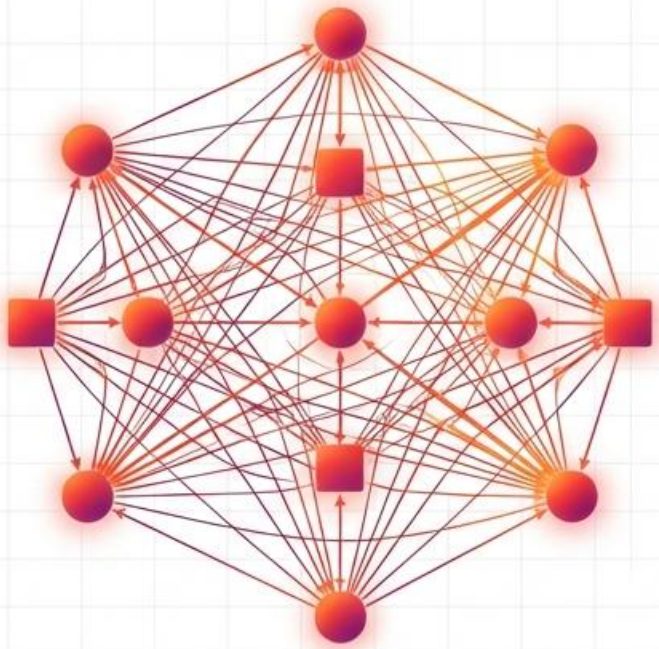
Astrocyte



Spatiotemporal Integration

← BIOMIMETIC PARALLEL →

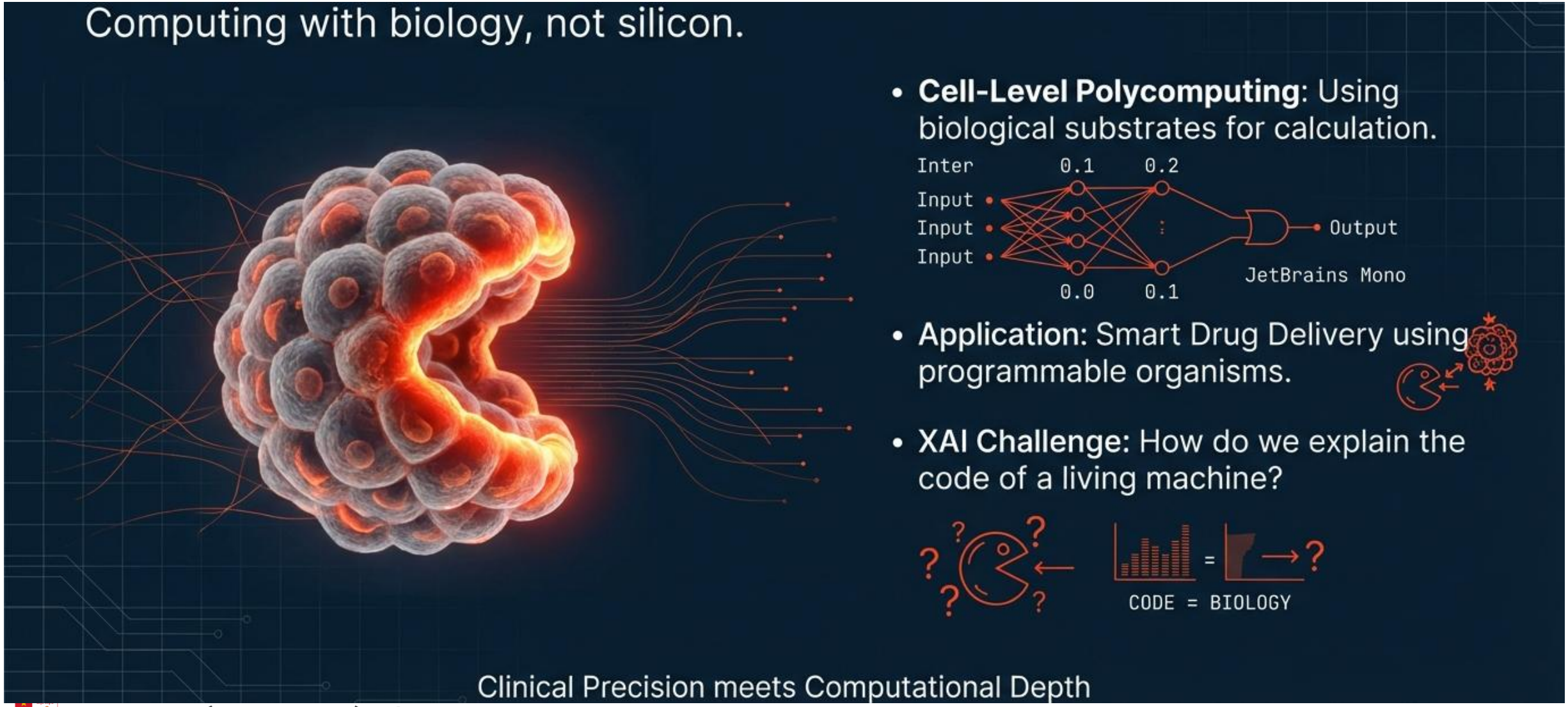
Transformer Attention Mechanism



Global Context Management

Clinical Precision meets Computational Depth

Polycomputing & Xenobots



Toward Human-Like Intelligence (HLI)

Clinical Surface:
Grad-CAM++ gives sight.

Neural Interface:
Saliency gives insight.

Cellular Depths:
Bio-AI gives life.

Clinical Precision meets Computational Depth

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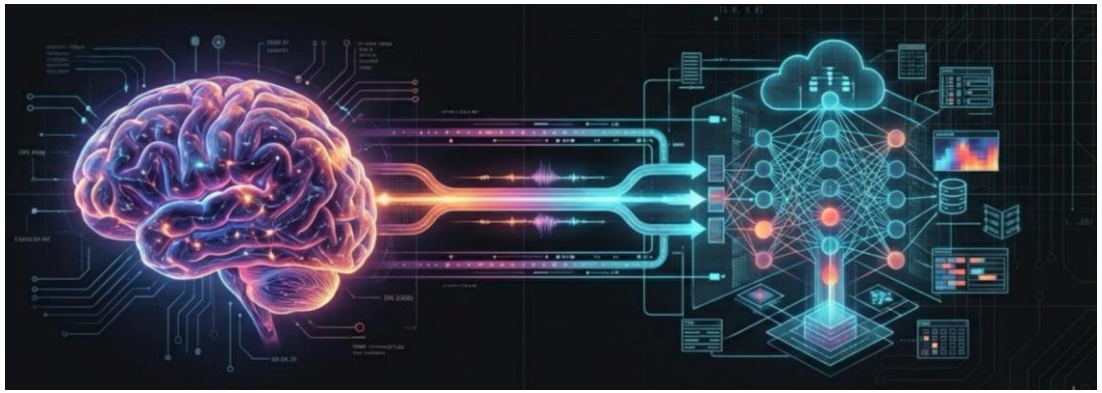
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